Correlation filter tracking

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I. Introduction

In this assignment, the short-term SiamFC tracker was adapted into a long-term tracking framework by using the peak value of the correlation response as an indicator of target presence. Various thresholds were tested to determine the optimal value for detecting target disappearance. The impact of the number of sampled regions on the tracker's re-detection performance was also evaluated. Cases where the tracker successfully re-detected the target were identified and visualized. For the re-detection mechanism, uniform sampling across the entire image was initially employed, followed by an evaluation of Gaussian sampling centered around the last confident target position.

II. Experiments

A. Short-term vs long-term tracker performance

In the first part of the evaluation, we began by testing the short-term implementation of the SiamFC tracker and then compared it to a modified long-term version. Both trackers were evaluated on all sequences in the provided dataset. The long-term modification builds on the short-term tracker by introducing a thresholding mechanism based on the correlation response. When the peak response falls below a predefined threshold, a re-detection step is triggered. This step samples 50 candidate positions uniformly across the entire image and selects the one with the highest correlation response. The re-detection process remains active until a response exceeding the threshold value of 4 is found.

The comparison of the trackers is presented in Table I. While the precision of the long-term variant decreased slightly—possibly due to re-detection being unnecessarily triggered in some frames where the target was still present, leading to suboptimal localization—the recall and, consequently, the F-score improved significantly. This improvement is expected, as the short-term tracker typically fails to recover once the target leaves the field of view, whereas the long-term extension is able to re-detect and continue tracking.

Tracker	Precision	Recall	F-score
Short-term	0.5982	0.3251	0.4212
Long-term	0.5870	0.4501	0.5095

Table I
Performance of short-term and long-term tracking.

B. Searching for the optimal value of confidence threshold

As previously mentioned, the maximum value of the correlation response was used as a confidence score. In the initial experiments, the threshold was set to 4 based on empirical observations: the correlation peak typically dropped from values around 6–9 to approximately 2–3 when the target was lost. The chosen threshold thus fell midway between these ranges. In this section, we systematically evaluate a range of thresholds around this preliminary value to identify the most effective setting.

On Figure 1 we can observe the results at different thresholds. We can see that the best overall F-score is achieved with the thresholds set to 4 or 5. For further evaluations we opt for 4 since this way we have less unnecesary re-detections. We can see that the precision actually keeps growing with a higher threshold, however the recall plummets.

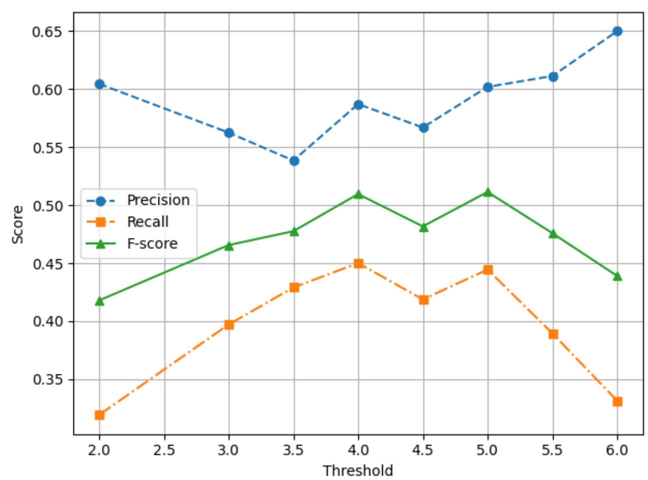

Figure 1. Comparison of performance at different thresholds

C. Testing different number of randomly sampled regions

Next we tested how the number of regions sampled during re-detection affects re-detection capability.

D. Visualization

Here we present the visualizations of re-identifications of targets using our tracker.

E. Different sampling methods

We also tried using the gaussian sampling method, around the last confidently tracked point (before the correlation output was below the threshold).

III. Conclusion